Phú
Hồ Nguyễn Phú

Leader: Hồ Nguyễn Phú
- Cao Tuệ Anh
- Phạm Trọng Quý
- Châu Minh Quân
- Hồng Nguyễn Phúc

Tutoring 3

1.

a) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0,08}{0,46} = \frac{4}{23}$

b) $P(C|A) = \frac{P(C \cap A)}{P(A)} = \frac{0,19}{0,38} = \frac{19}{32}$

c) $P(B|A \cap B) = \frac{P(B \cap (A \cap B))}{P(A \cap B)}$

$= \frac{P(A \cap B)}{P(A \cap B)} = 1$

d) $P(B|A \cup B) = \frac{P(B \cap (A \cup B))}{P(A \cup B)}$

$= \frac{P(B)}{P(A \cup B)} = \frac{0,46}{0,7} = \frac{23}{35}$

e) $P(A|A \cup B \cup C) = \frac{P(A \cap (A \cup B \cup C))}{P(A \cup B \cup C)}$

Thứ ..... ngày ..... tháng ..... năm .....

Phu

Hồ Nguyễn Phú

$$= \frac{P(A)}{P(A \cup B \cup C)} = \frac{0,32}{0,88} = \frac{4}{11}$$

f) $$P(A \cap B \mid A \cup B) = \frac{P((A \cap B) \cap (A \cup B))}{P(A \cup B)}$$

$$= \frac{P(A \cap B)}{P(A \cup B)} = \frac{0,08}{0,7} = \frac{4}{35}$$

2) Let A be the event that an order whose value is over 1000 \$ and B be the event that customer will pay with credit card.

According to the given statistics:

$P(A) = 0,42$

$P(B|A) = 0,63$

a)

Let A_1, A_2, A_3 be the event that the first, second, and third order be valued at over 1000 \$, respectively

A_1 and A_2 are independent.

$\Rightarrow P(A_2|A_1) = P(A_2)$

Since A_1 occurring does not change the probability value of A_2.

Therefore:

$$P(A_1 \cap_2) = P(A_1) . P(A_2|A_1)$$
$$= P(A_1) . P(A_2)$$

The same goes for A_3.

In conclusion:

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) . P(A_2) . P(A_3)$$
$$= 0,42 . 0,42 . 0,42$$
$$= 0,074$$

Thứ ..... ngày ..... tháng ..... năm .....

b) $P(\overline{B}|A) = 1 - P(B|A)$

$= 1 - 0,63 = 0,37$

$P(\overline{B} \cap A) = P(A) \cdot P(\overline{B}|A)$

$= 0,42 \cdot 0,37$

$= 0,1554$

Phú
Hồ Nguyễn Phú

5) Let:

X: card from red suit

Y: card from black suit

a) $P(A\heartsuit \mid X) = \dfrac{P(A\heartsuit)}{P(X)} = \dfrac{1/52}{1/2} = \dfrac{1}{26}$

b) $P(\heartsuit \mid X) = \dfrac{P(\heartsuit)}{P(X)} = \dfrac{13/52}{1/2} = \dfrac{1}{2}$

c) $P(Y) = \dfrac{P(\spadesuit)}{\cancel{52}} + \dfrac{P(\clubsuit)}{\cancel{52}} = \dfrac{1}{2}$

d) $P(\heartsuit \mid Y) = \dfrac{P(\heartsuit \cap Y)}{P(Y)} = 0$

~~e) $P(K \mid X)$~~

e) $P(K \mid X) = \dfrac{P(K \cap X)}{P(X)} = \dfrac{2/52}{1/2} = \dfrac{1}{13}$

f) $P(K \mid \text{red picture card}) = \dfrac{P(K \cap \text{red picture card})}{P(\text{red picture card})}$

$= \dfrac{2/52}{6/52} = \dfrac{1}{3}$

Phu
Hồ Nguyên Phú

4. Let A : blue balls
A' : red balls
S : shiny
S' : dull

According to the given statistics

$$P(A) = \frac{54}{150} = \frac{9}{25}$$

$$\Rightarrow P(A') = 1 - P(A) = 0{,}64$$

and

$$P(A' \cap S) = \frac{36}{150} = \frac{6}{25}$$

4.1) $$P(S|A') = \frac{P(S \cap A')}{P(A')} = \frac{6/25}{0{,}64} = \frac{3}{8}$$

4.2) $$P(S'|A') = 1 - P(S|A')$$
$$= 1 - \frac{3}{8} = \frac{5}{8}$$

5) Let

A: pass on apperance
P: pass on performance
C: pass on cost

According to the given statistics:

Thứ ..... ngày ..... tháng ..... năm .....

Phu
Hồ Nguyễn
Phú

$P(A \cap C) = 0,4$

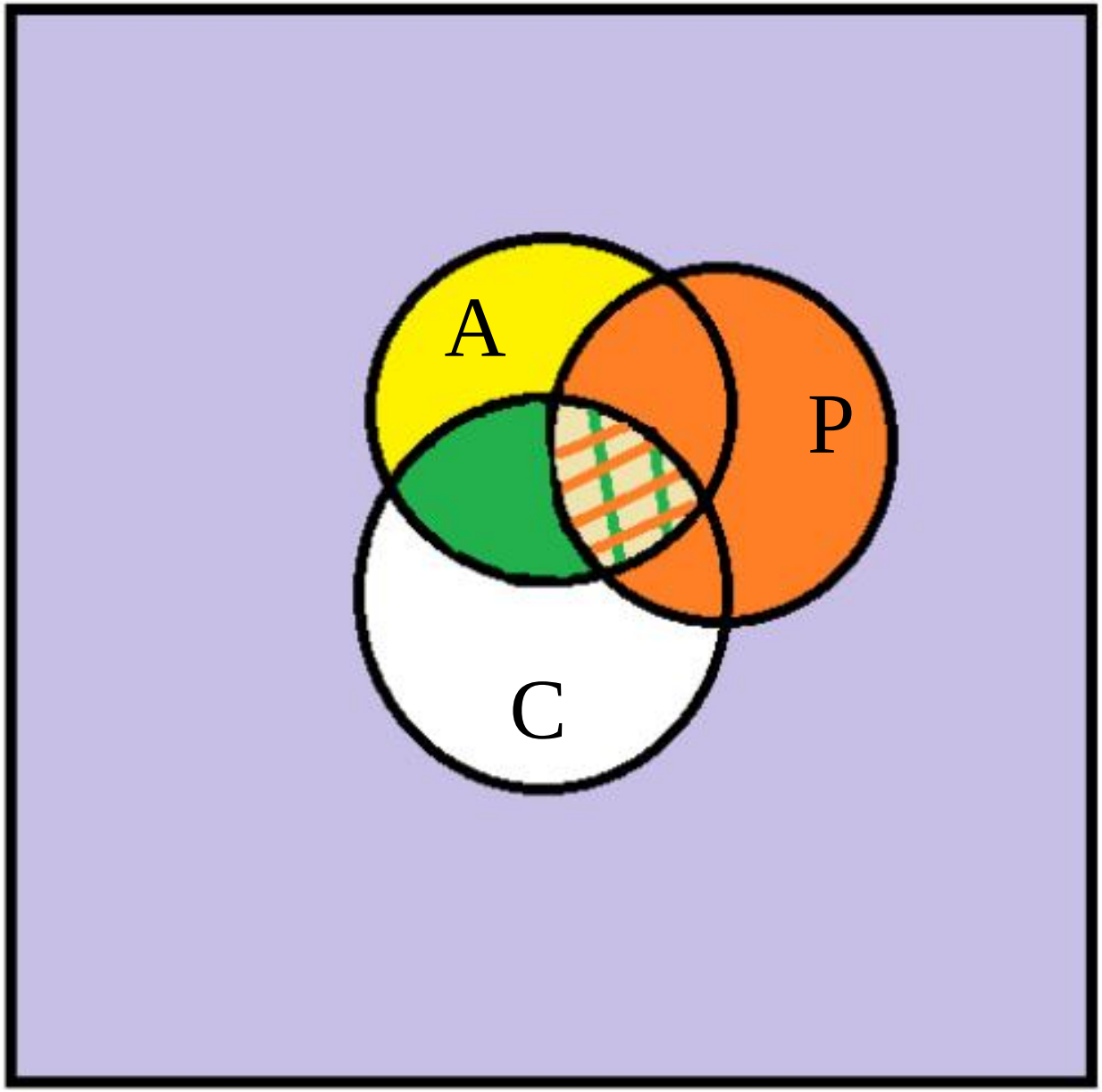
$P(A \cap P \cap C) = 0,31$

$P(P) = 0,64$

$P(A' \cap P' \cap C') = 0,19$

$P(A \cap P' \cap C') = 0,06$

We have a diagram:



$A \cap C$

P

$A \cap P' \cap C'$

$A' \cap P' \cap C'$

$A \cap P \cap C$

Phan
Hồ Nguyễn
Phú

a) $P(C \cap P' \cap A') = 1 - P(A' \cap P' \cap C') + P(A \cap P \cap C)$
$- P(P) - P(A \cap C) - P(A \cap P' \cap C')$
$= 1 - 0{,}19 + 0{,}31 - 0{,}64 - 0{,}4 - 0{,}06$
$= 0{,}02$

b) $P((A \cap C \cap P) \mid (A \cap C))$
$= \dfrac{P(A \cap C \cap P)}{P(A \cap C)} = \dfrac{0{,}31}{0{,}4} = \cancel{0{,}075}\ 0{,}775$